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(9630) PH03

Report on the examination

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REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) UNIT PH03 JANUARY 2023

The range of question styles and variety of topics in Section A gave students many opportunities to demonstrate their knowledge and understanding. As has been mentioned before in these reports, mathematical questions were answered well by many of the students. An important point to stress is that, even when the final answer was incorrect, students who set out their answer clearly often obtained some marks for a clear method. Most students continued to demonstrate difficulties when answering questions that required a written response. Answers failed to get marks because they were vague or incomplete, or did not answer the question.

SECTION A

QUESTION 01

Most students drew the correct arrow in 01.1. The most common error was to draw an arrow in the direction the bucket is moving. Multiple and careless arrows received no credit.

Answers to 01.2 highlighted the very common misconception that the centripetal force is an extra force rather than a combination of the forces already acting. At other points in the bucket's motion the centripetal force on the water is a combination of its weight and the reaction force from the bucket. In 01.2 it is only the weight, as the reaction from the bucket is zero. This answer was rarely seen.

The calculation in 01.3 was answered correctly by the majority of students. Most who made any progress went on to get full marks.

Although many students were able to give two correct changes in 01.4, very few made any effort to discuss or provide any reasoned support for their suggestions. This limited the marks awarded.

QUESTION 02

Most answers to 02.1 included the definition of simple harmonic motion. Many fewer went on to link this to the properties of the graph. Only the best answers referred to the fact that the line passed through the origin.

The 'show that' calculation in 02.2 produced a range of marks and discriminated very well. Most students completed the calculation without problem, and provided an answer to more than one significant figure.

The calculation in 02.3 also discriminated very well. It was pleasing to note that many students made use of the 'show that' value from 02.2 in their calculation. Students should be encouraged to do this even when they were unable to obtain the 'show that' value in the earlier part.

The best answers to 02.4 placed a very clear label at the origin of Figure 4. Descriptive answers were not credited.

QUESTION 03

The mixture of demand in the different parts produced a very good range of marks in this question.

03.1 was quite difficult although only a few students erroneously thought that the suggestion was correct. The best answers made a reference to the relationship between electric field strength E and electric potential. Common incorrect answers suggested that the field was uniform and those students clearly did not understand that the graph would then have been a horizontal line. Other incorrect answers were based on the idea that E was the potential.

The calculation in 03.2 was easier, with most students getting full marks. Some marks were available for incorrect answers when correct steps were identified by the examiner.

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Many students were able to make some progress in 03.3 but full marks were rarely awarded. The most common mark given was the second marking point for stating that the horizontal component of the acceleration decreased. For the first marking point, the relationship between the field strength and force was needed. Many students made no reference to the direction and therefore made no attempt at the third point.

It had been made clear in the question that the acceleration was not constant. Despite this, many students attempted a constant-acceleration solution to the calculation in 03.4. This could lead to the correct numerical answer but the physical principles behind the solution are incorrect; this route was not accepted. The best answers set out each step in this energy calculation to obtain the correct answer.

QUESTION 04

The first two question parts indicated the difference between students' ability to do mathematical problems and their facility with written explanations. The straightforward calculation in 04.1 was answered correctly by nearly 90% of students. The explanation in 04.2 was answered correctly by less than 10% of students.

A major misconception seen in answers to 04.2 was the idea that the '*resistance*' of the capacitor increases as it is charged. Other answers included the idea that the current decreased, despite the question making it clear that it was constant.

04.3 was usually answered well with most students realising that, as it was a 'show that' question, the equation being used had to be seen. It should be noted that making clear the principles being used is always good practice whether or not the question is a 'show that'.

Most students had no problem correctly answering 04.4. About 10% of students received one mark for an incorrect answer provided the reading from the graph and the use of the equation was seen. Several students used the gradient approach, which was also given credit.

04.5 was also answered correctly by most students. A surprisingly large number omitted the permittivity of free space, however. This question discriminated very well.

The discussion in 04.6 challenged most students. The most common mark awarded was for stating that the capacitance of the new capacitor was much smaller. Many students then made no attempt to answer the question but described the effect of the dielectric. Few students discussed the experimental procedure and many of these incorrectly suggested that the charging time would be longer. References to differences in voltage reached were also incorrect. Although a numerical solution was not expected, some students obtained full marks after calculating a charging time and stating that this would be too short to be measured using a stop watch.

QUESTION 05

In order to get the mark in 05.1, students had to state clearly what was being halved. Some incorrect answers only stated that it was the fluorine-18, and others referred to the fluorine-18 nuclei halving. Students should be encouraged to learn a standard definition of half-life that refers to mass, number of nuclei or activity for example.

05.2 was answered correctly by most students, with only a few apparently correct calculations not being rewarded due to missing working.

The calculation in 05.3 provided a good range of marks and discriminated very well. Many students had difficulties relating the mass of one mole of fluorine to the number of fluorine atoms.

A common error in 05.4 was the confusion between the activity A and the initial activity A_o. This suggests that some students know how to do the mathematics, but do not know how to apply it to the information in the question.

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QUESTION 06

In 06.1, a surprising number of students incorrectly suggested that the particle moves at right angles to the field to experience no force. Other students had difficulty expressing the direction of the velocity relative to the field. The use of the word '*parallel*' was not required but was the clearest way of answering the question.

06.2 discriminated very well. The most common error was to use a resultant velocity rather than the perpendicular component in the calculation.

Many students obtained a mark in 06.3 by showing some understanding that the path was a combination of linear and circular motion. Far fewer were able to explain this using correct physics.

Many students were able to give some reference to Faraday's Law, although some did not include a reference to the '*rate of change of flux linkage*', and just mentioned '*flux*'. Applying the law to this situation was one of the most demanding things on the paper. Some students understood the importance that the change was rapid, but very few understood that the very large area was related to the very long power lines.

QUESTION 07

Most parts of this question discriminated very well.

The 'show that' calculation in 07.1 was answered correctly by most students although some lost the mark because their working was not sufficiently clear.

In 07.2 the best answers made use of the blank answer space to do a calculation that showed that the data were not consistent with circular motion.

The calculation in 07.3 was completed correctly by many students. The easiest way to approach this problem was to give the period of the Earth the value of 1 (year) and its orbital radius the value of 1 (AU). It was more common to see answers that dealt with seconds and metres, however. This approach led to several arithmetic errors in the answers that were seen.

In 07.4 there were several ways to obtain the first marking point. The correct answer was needed for the second mark point and this was rarely seen. Students who gave an answer to one significant figure did not get this mark, for example.

Although there were several answers that appeared to have some idea of what was needed for 07.5, the answers were often poorly expressed and ambiguous; the marks were rarely given.

SECTION B

Most of the students answered more than half of the multiple-choice questions correctly. The easiest questions were 8, 9, and 15.

In all of the questions the most popular answer was the correct answer. The questions with significant distractors were 10, 12 and 18.

In question 10, over one third of students chose option D. Some of these students may not have related the maximum gravitational potential to the zero gravitational field strength.

In question 12 about one quarter of students chose option C. This may be because object P was the closest to X.

In question 18 over one third of students chose option D. This may indicate an arithmetic error associated with sines and cosines.

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